

TAMIZHAZHAGAN S

ECE Engineer · Full-Stack Java Developer · IoT Systems Researcher · Embedded Systems

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01 — PROFILE

B.E. ECE graduate (Anna University, 2023–2026) who builds systems at the boundary of **hardware and human safety**. **Peer-reviewed research published in IJEETR (Mar–Apr 2026)** — a life-critical IoT helmet with sub-5s emergency response — alongside hands-on Java internship experience and production-grade full-stack development. Writes **100Hz embedded C** the same week he architects SQL schemas and clears LeetCode Hard problems. 12 verified HackerRank certifications, 54+ LeetCode problems solved. Actively seeking roles in embedded systems, IoT engineering, or full-stack Java development.

12+

CERTIFICATIONS

54+

LEETCODE SOLVED

×1

PUBLISHED PAPER

4+

REAL PROJECTS

6

LANGUAGES

<5s

IOT LATENCY

02 — INTERNSHIP EXPERIENCE

Full Stack Development Intern · Industry Internship

Sep – Oct 2025

- Designed and developed responsive web applications integrating **HTML/CSS/JS** frontend with **Java & MySQL** backend using RESTful API principles and OOP design patterns.
- Optimised SQL queries and UI performance; delivered sprint-cycle features using **Git-based** collaborative workflows with 100% on-time delivery.
- Maintained comprehensive technical documentation and weekly status reports across the full SDLC; collaborated cross-functionally with design and backend teams.

Telecom & Embedded Systems Intern · Industry Internship

Feb 2025

- Worked on **PCB design**, telecom circuit layout, and embedded systems testing across communication network infrastructure.
- Debugged hardware-software interface issues; produced structured test observation and compliance records.

Java Development Intern · Industry Internship

2023

- Gained production-level Java development experience with **JDBC** database integration and **OOP** design patterns on a real codebase.
- Implemented professional workflows: version control, code review, and structured project delivery.

03 — RESEARCH & PUBLICATION

IoT-Based Smart Helmet — Accident Detection & Emergency Alert System

IJEETR · Mar–Apr 2026 · Peer-Reviewed & Published

Designed a life-critical, fully offline embedded system achieving **sub-5s end-to-end emergency response** for motorcycle accident victims — zero internet dependency.

```
// pipeline
MPU6050 100Hz IMU monitor → Crash-threshold calibration → NEO-6M GPS fetch → SIM800L SMS alert
```

- 100Hz IMU sampling with crash threshold calibration — eliminates false positives.
- ₹3,500 BOM cost — deployable at population scale; production-ready hardware prototype.
- Led a team of 4 across a 6-month development cycle: requirements, firmware, FMEA analysis, prototype testing, and peer-reviewed technical documentation.

BookMyShow Ticket Booking System

Java · JDBC · MySQL · OOP · MVC

- Full-stack Java desktop app with dual-role authentication (admin + user) and role-based access control.
- JDBC-backed transactional seat locking with **SELECT FOR UPDATE** — eliminates double-booking race conditions.
- Clean MVC architecture with zero code duplication; robust SQL schema enforcing referential integrity.

Smart Appointment Scheduler

HTML · CSS · JavaScript · MySQL · Full-Stack

- Full-stack web application with scheduling, CRUD operations, and a reminder system.
- Responsive UI with dynamic reminders and real-time slot availability updates.
- Normalised MySQL schema with relational integrity and clean separation of business/data logic.

Automatic Industrial Cleaning Bot

Robotics · Embedded C · Ultrasonic · Motor Control · Wireless RF

- Autonomous mobile robot with ultrasonic obstacle detection and differential drive control.
- Wireless remote override and repeatable cleaning path logic — **3rd Prize**, Project Expo, Muthayammal College (Dec 2024).

FM Spy Bug — Wireless FM Transmitter

RF Electronics · Analog Circuit Design · KiCad PCB

- Discrete-component FM transmitter — no microcontroller — with optimised oscillator and impedance matching.
- Full KiCad schematic to fabrication-ready PCB layout; clean signal on standard FM receivers.

05 — SKILLS

LANGUAGES

Java · C · Embedded C · Python · JavaScript · SQL

WEB / BACKEND

HTML5 · CSS3 · React.js · Node.js · JDBC · REST APIs · MySQL · OOP · MVC

EMBEDDED / IOT

Arduino Nano/Uno · MPU6050 · NEO-6M GPS · SIM800L · AT Commands · Sensor Fusion

EDA / TOOLS

KiCad · Proteus · Multisim · Eagle · Arduino IDE · VS Code · IntelliJ · Git · Linux

PROBLEM SOLVING

LeetCode 54+ (incl. Hard) · HackerRank 12 Certs · DSA & Algorithms

06 — EDUCATION

B.E. — Electronics & Communication Engg.

2023 — 2026

AVS Engineering College, Anna University, Salem

Diploma — Electrical & Electronics Engg.

2020 — 2023

Thiagarajar Polytechnic College, Salem

SSLC

2019 — 2020

Govt. Higher Secondary School, Singipuram

07 — CERTIFICATIONS

★ Software Engineer (Full-Stack) — Verified

Software Engineer Intern

Java (Basic)

JavaScript (Basic & Intermediate)

Python (Basic)

CSS (Basic)

SQL (Basic · Intermediate · Advanced)

Problem Solving (Basic & Intermediate)

Cloud Infrastructure & AI — GUVU/IIT Madras

Java Programming — Livewire

PCB Design & Simulation

ML Basics — GUVU

08 — ACHIEVEMENTS

- ▶ **Published Researcher** — IJEETR, Mar–Apr 2026 · Peer-reviewed, accepted, indexed.
- ▶ **3rd Prize** — Project Expo "Clean Energy Projections," Muthayammal College (Dec 2024).
- ▶ **Technical Presenter** — "Smart Cleaning Bot," VARNAM'24, KIT Engineering College (Dec 2023).
- ▶ 54+ LeetCode problems solved, including Hard tier · 12 Verified HackerRank Certifications.

09 — SOFT SKILLS

Analytical Thinking & Problem Solving

Cross-functional Collaboration

Leadership — led 4-member research team

Adaptability & Self-driven Learning

Professional Written & Verbal Communication